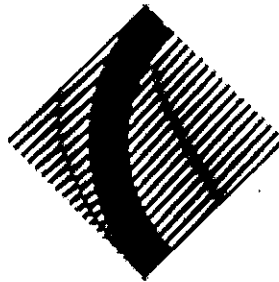


MK
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EW
AB
GB

NAME _____

CLASS 11MTX _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2017

YEAR 11

AP2

MATHEMATICS EXTENSION 1

Time allowed – 1½ HOURS

Plus 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- Write your name on the question paper.
- Multiple choice questions should be answered on the separate multiple choice answer sheet.
- Each question in Section II is to be commenced in a new booklet clearly marked with your name, class and the relevant question number.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used and the Reference Sheet is provided.

Section I (10 marks)

Attempt Questions 1–10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1 – 10.

1. If $\cos \alpha = \frac{1}{3}$, where α is an acute angle, find the value of $\cos 2\alpha$.

A. $-\frac{7}{9}$

C. $\frac{2}{3}$

B. $-\frac{1}{3}$

D. $\frac{2}{9}$

2. Simplify $\frac{(n+1)! + (n+2)!}{(n+1)!}$

A. $1 + (n+2)!$

C. $n+3$

B. $n+2$

D. $(n+2)!$

3. The solution of $\frac{1}{4x} > x$ is:

A. $x < 0$ or $x > \frac{1}{2}$

B. $-\frac{1}{2} < x < \frac{1}{2}$

C. $x < -\frac{1}{2}$ or $x > \frac{1}{2}$

D. $x < -\frac{1}{2}$ or $0 < x < \frac{1}{2}$

4. Point P internally divides the interval between $(2, 4)$ and $(4, 8)$ in the ratio 2: 1. What are the coordinates of P ?

A. $\left(2\frac{2}{3}, 5\frac{1}{3}\right)$

B. $(3, 12)$

C. $\left(3\frac{1}{3}, 6\frac{2}{3}\right)$

D. $(4, 6)$

5. Correct to the nearest degree, what is the size of the acute angle between the lines $3x - 2y = 0$ and $x + 2y = 0$?
- A. 30°
 - B. 48°
 - C. 65°
 - D. 83°
6. If $5^x = 17$, then x is closest in value to:
- A. 1.760
 - B. 1.761
 - C. 0.567
 - D. 0.568
7. When compared to the graph of $y = f(x)$, the graph of $y = f(2 - x)$ can be created by:
- A. Reflecting $f(x)$ vertically, then shifting it 2 units to the right
 - B. Reflecting $f(x)$ vertically, then shifting it 2 units to the left
 - C. Reflecting $f(x)$ horizontally, then shifting it 2 units to the right
 - D. Reflecting $f(x)$ horizontally, then shifting it 2 units to the left
8. The coefficient of x^2 in the expansion of $\left(2x + \frac{3}{x}\right)^6$ is:
- A. 135
 - B. 576
 - C. 2160
 - D. 4860

-

A. 30°
B. 35°

C. 40°
D. 70°

-

A. 3.5 C. 8.0
B. 4.5 D. 12.5

Section II (48 marks)**Attempt Questions 11–14****Allow about 1 hour 15 minutes for this section.**

Answer each question in the writing booklets provided. Begin each question in a new booklet.

Question 11 (12 marks) – START A NEW WRITING BOOKLET		Marks
(a)	α and β are the roots of $4x^2 - 2x - 5 = 0$. Without evaluating α and β individually, find the exact values of:	
	(i) $\alpha + \beta$	1
	(ii) $\alpha\beta$	1
	(iii) $ \alpha - \beta $	2
	(iv) $\alpha - 2\alpha^2$	1
(b)	Consider the function $y = x^2 - 2(k + 1)x - k + 5$.	
	(i) Show that the discriminant is $4(k^2 + 3k - 4)$.	1
	(ii) Hence or otherwise find the values of k such that the function is indefinite.	2
(c)	A piece of wire 12cm long is cut into two pieces. The first length is bent to form a square, while the second length is bent to form a rectangle whose length is 3 times its breadth.	
	(i) Let the side length of the square be x cm and the breadth of the rectangle be y cm. Show that $x + 2y = 3$.	1
	(ii) Find the dimensions of the square such that the sum of the areas, A cm ² , is the smallest possible value.	3

Question 12 (12 marks) – START A NEW WRITING BOOKLET

Marks

- (a) (i) State the domain of $y = \frac{x-1}{x(x+2)}$. 1

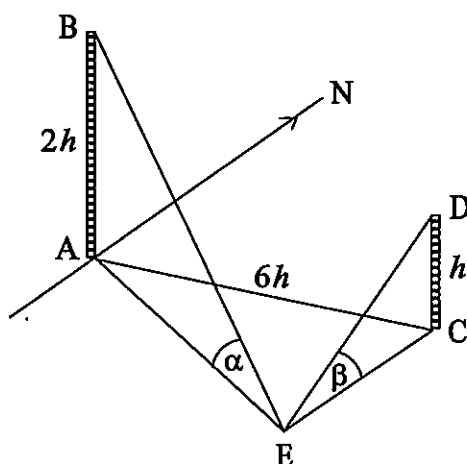
- (ii) Hence, graph $y = \frac{x-1}{x(x+2)}$. 3

Clearly indicate all intercepts & asymptotes with their equations.

- (b) A and C are the bases of two buildings AB and CD with heights $2h$ metres and h metres respectively. 3

CD is due east of AB and at a distance of $6h$ metres from AB .

From a point E due south of CD , the angles of elevations of AB and CD are α and β respectively.



Show that $4 \cot^2 \alpha - \cot^2 \beta = 36$.

- (c) Consider the curve $y = x^4 - 2x^2 - x$.
- (i) Find the equation of line l , the tangent at $(1, -2)$. 2
- (ii) Show that line l is also tangent to the curve at another point, and find the coordinates of this point. 3

Question 13 (12 marks) – START A NEW WRITING BOOKLET**Marks**

- (a) M is the midpoint of hypotenuse AC in a right-angled triangle, $\triangle ABC$. Draw a neat diagram showing this information and hence prove that $BM = CM$. **3**
- (b) (i) If $f(x) = \log_3(x^2)$, evaluate $f(-1)$. **1**
(ii) Neatly sketch the graph of $y = f(x)$, showing all features. **3**
- (c) Show that $\frac{\tan^3 \theta + 1}{\tan \theta + 1} = \frac{1}{2} \sec^2 \theta (2 - \sin 2\theta)$ **3**
- (d) x and y are rational numbers such that: **2**
- $\sqrt{x + y}$ is irrational
 - $x - 2y + \sqrt{x + y} = \frac{1}{2}\sqrt{6}$

Find the values of x and y .

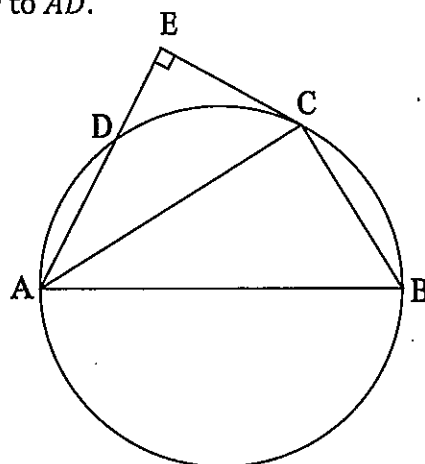
Question 14 (12 marks) – START A NEW WRITING BOOKLET

- (a) Consider the function $E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.
- (i) Write out, in simplified form, the first 5 terms of $E(x)$. **1**
- (ii) Hence prove that $E'(x) = E(x)$. **1**
- (b) By using the auxiliary angle method, solve $\cos 2x + \sqrt{3} \sin 2x = \sqrt{2}$ in the domain $0 \leq x \leq 360^\circ$. **3**

Question 14 continues on the next page

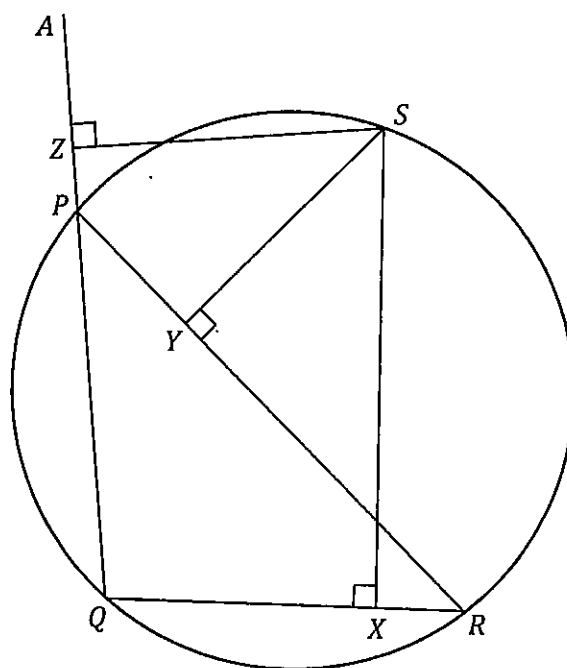
- (c) A, B, C and D are points on a circle with diameter AB .
The tangent at C meets the chord AD produced to E such that CE is perpendicular to AD .

3



Show that AC bisects $\angle EAB$.

- (d) P, Q, R and S are points on the circumference of a circle. SX, SY and SZ have been constructed perpendicular to QR, PR and PQ as shown in the diagram below.



Neatly copy this diagram into your writing booklet.

- (i) Prove that P, S, Y and Z are concyclic.

2

- (ii) Prove that R, S, X and Y are concyclic.

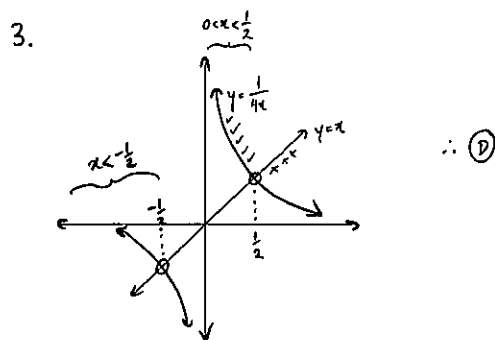
2

End of Examination

Section I

$$\begin{aligned} 1. \cos 2x &= 2\cos^2 x - 1 \\ &= 2\left(\frac{1}{3}\right)^2 - 1 \\ &= \frac{2}{9} - 1 \\ &= -\frac{7}{9} \therefore \textcircled{A} \end{aligned}$$

$$\begin{aligned} 2. \frac{(n+1)! + (n+2)!}{(n+1)!} &= \frac{(n+1)! + (n+2)(n+1)!}{(n+1)!} \\ &= 1 + (n+2) \\ &= n+3 \therefore \textcircled{C} \end{aligned}$$



$$\begin{aligned} 4. P &= \left(\frac{kx_1 + lx_2}{k+l}, \frac{ky_1 + ly_2}{k+l} \right) \\ &= \left(\frac{1(2) + 2(4)}{3}, \frac{1(4) + 2(8)}{3} \right) \\ &= \left(\frac{11}{3}, \frac{20}{3} \right) \therefore \textcircled{C} \end{aligned}$$

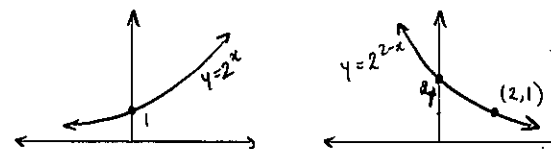
2017 Year 11 Mathematics Extension 1 AP2 Solutions

$$\begin{aligned} 5. \quad 3x - 2y &= 0 \rightarrow y = \frac{3}{2}x \\ x + 2y &= 0 \rightarrow y = -\frac{1}{2}x \\ \tan \theta &= \left| \frac{\frac{3}{2} - (-\frac{1}{2})}{1 + (\frac{3}{2} \times -\frac{1}{2})} \right| \\ &= \frac{2}{1 - \frac{3}{4}} \\ \tan \theta &= 8 \end{aligned}$$

$$\theta = 82.87... \therefore \textcircled{D}$$

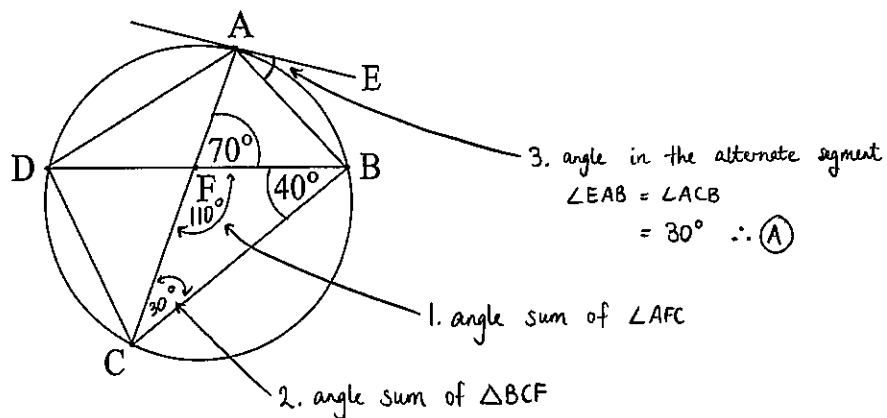
$$\begin{aligned} 6. \quad 5^x &= 17 \\ x &= \log_5 17 \\ &= \frac{\log 17}{\log 5} \\ &= 1.76037... \therefore \textcircled{A} \end{aligned}$$

$$\begin{aligned} 7. \quad f(2-x) &= f(-x+2) \\ &= f[-(x-2)] \therefore \textcircled{C} \\ \text{eg. Compare } y &= 2^x \text{ with } y = 2^{2-x} \end{aligned}$$



$$\begin{aligned} 8. \text{ For } \left(2x + \frac{3}{x}\right)^6, \text{ the general term } T_n &= \binom{6}{n} (2x)^n \left(\frac{3}{x}\right)^{6-n} \\ &= \binom{6}{n} 2^n x^n 3^{6-n} x^{n-6} \\ &= \binom{6}{n} 2^n 3^{6-n} x^{2n-6} \\ \therefore x^2 \text{ term is } T_4 &= \binom{6}{4} 2^4 3^2 x^2 \\ &= 2160x^2 \therefore \textcircled{C} \end{aligned}$$

9.



10. $AP \times BP = PC^2$ (square of intercept on tangent equals product of intercepts on secant)
 $(AB+8) \times 8 = 10^2$

$$AB + 8 = 12.5$$

$$AB = 4.5 \therefore \text{(B)}$$

Section II

QUESTION 11

(a) $4x^2 - 2x - 5 = 0$

(i) $\alpha + \beta = \frac{-(-2)}{4}$

$$= \frac{1}{2}$$

(ii) $\alpha\beta = \frac{-5}{4}$

(iii) $|\alpha - \beta| = \sqrt{(\alpha - \beta)^2}$
 $= \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$
 $= \sqrt{\frac{1}{4} + 5}$
 $= \sqrt{\frac{21}{4}}$
 $= \frac{\sqrt{21}}{2}$

① manipulation of sum/product results

① final value

(iv) METHOD 1: From (i), $\beta = \frac{1}{2} - \alpha$
 USING SUM & PRODUCT RESULTS From (ii), $\beta = \frac{-5}{4\alpha}$

Equating gives $\frac{1}{2} - \alpha = \frac{-5}{4\alpha}$
 $\frac{4\alpha}{2} - 4\alpha^2 = -5$
 $2\alpha - 4\alpha^2 = -5$
 $\alpha - 2\alpha^2 = \frac{-5}{2}$

METHOD 2:
 USING ORIGINAL EQUATION

α is a root of $4x^2 - 2x - 5 = 0$.

$$\therefore 4\alpha^2 - 2\alpha - 5 = 0$$

$$4\alpha^2 - 2\alpha = 5$$

$$2\alpha^2 - \alpha = \frac{5}{2}$$

$$\alpha - 2\alpha^2 = \frac{-5}{2}$$

$$(b) y = x^2 - 2(k+1)x - k + 5$$

$$(i) \Delta = [-2(k+1)]^2 - 4(1)(-k+5)$$

$$= 4(k^2 + 2k + 1) + 4(k - 5)$$

$$= 4(k^2 + 2k + k + 1 - 5)$$

$$= 4(k^2 + 3k - 4) \text{ as required.}$$

(ii) Function is indefinite when $\Delta \geq 0$.

$$\text{ie. } 4(k^2 + 3k - 4) \geq 0$$

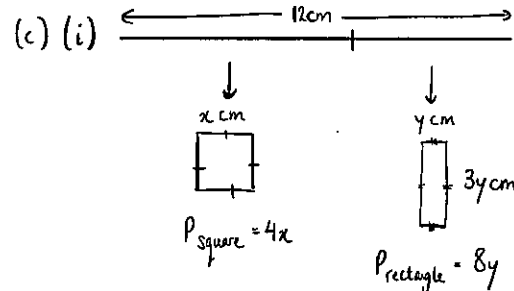
$$k^2 + 3k - 4 \geq 0$$

$$(k+4)(k-1) \geq 0$$

$$\therefore k \leq -4 \text{ or } k \geq 1.$$

① Correct values for k

① Student makes progress in solving $\Delta \geq 0$



$$\therefore 4x + 8y = 12$$

$$x + 2y = 3 \text{ as required.}$$

(ii)

$$x = 3 - 2y$$

$$\therefore A = \text{Square} + \text{Rectangle}$$

$$= (3 - 2y)^2 + (y \times 3y)$$

$$= 9 - 12y + 4y^2 + 3y^2$$

$$= 7y^2 - 12y + 9$$

① expression for unsimplified total area

This is a concave-up parabola, so its minimum value

occurs along the axis of symmetry:

① reasoning to locate minimum

$$y = \frac{-(-12)}{2(7)}$$

$$y = \frac{6}{7}$$

$$\therefore \text{Side length of square is } 3 - 2\left(\frac{6}{7}\right) = \frac{9}{7} \text{ cm.}$$

① side length

(ii) Function is indefinite when $\Delta \geq 0$.

$$\text{ie. } 4(k^2 + 3k - 4) \geq 0$$

$$k^2 + 3k - 4 \geq 0$$

$$(k+4)(k-1) \geq 0$$

$$\therefore k \leq -4 \text{ or } k \geq 1.$$

① Student makes progress in solving $\Delta \geq 0$

① correct values for k

(c) (i)

$$P_{\text{square}} = 4x$$

$$P_{\text{rectangle}} = 8y$$

$$\therefore 4x + 8y = 12$$

$$x + 2y = 3 \text{ as required.}$$

(ii)

$$x = 3 - 2y$$

$$\therefore A = \text{Square} + \text{Rectangle.}$$

$$= (3 - 2y)^2 + (y \times 3y)$$

$$= 9 - 12y + 4y^2 + 3y^2$$

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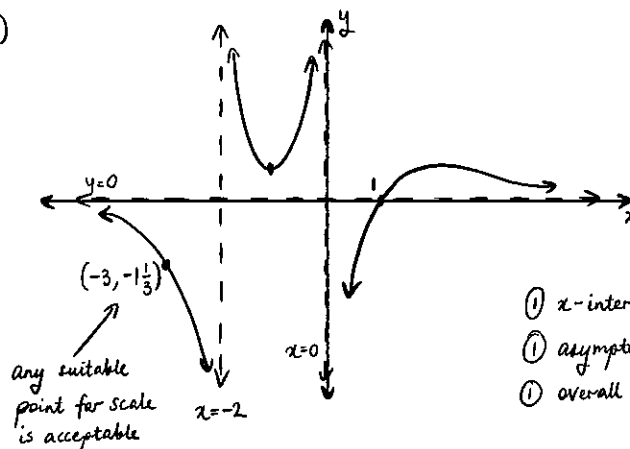
① side length

QUESTION 12

$$(a) y = \frac{x-1}{x(x+2)}$$

(i) Domain is $x < -2$, $-2 < x < 0$ or $x > 0$.

(ii)

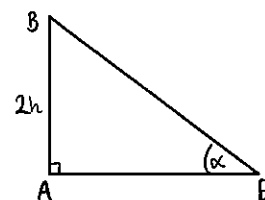


① x-intercept

① asymptotes

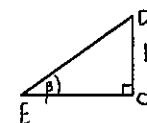
① overall shape

(b)



$$\text{In } \triangle ABE, \tan \alpha = \frac{2h}{AE}$$

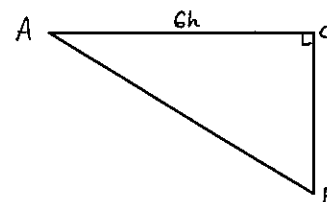
$$AE = 2h \cot \alpha$$



$$\text{In } \triangle CDE, \tan \beta = \frac{h}{CE}$$

$$CE = h \cot \beta$$

① expressions for AE & CE



In $\triangle ACE$, $AC^2 + CE^2 = AE^2$ (Pythagoras' Theorem)

$$(6h)^2 + (h \cot \beta)^2 = (2h \cot \alpha)^2$$

① substitution into Pythagoras

$$36h^2 + h^2 \cot^2 \beta = 4h^2 \cot^2 \alpha$$

$$36 + \cot^2 \beta = 4 \cot^2 \alpha$$

① algebraic manipulation

$$4 \cot^2 \alpha - \cot^2 \beta = 36 \text{ as required.}$$

$$(c) y = x^4 - 2x^2 - x$$

$$(i) \frac{dy}{dx} = 4x^3 - 4x - 1$$

$$\text{When } x=1, \frac{dy}{dx} = 4 - 4 - 1$$

$$= -1 \quad \textcircled{1} \text{ gradient at } (1, -2)$$

$$\therefore \text{Equation of } l \text{ is } y - (-2) = -1(x - 1)$$

$$y + 2 = -x + 1 \quad \textcircled{1} \text{ equation of line}$$

$$y = -x - 1 \quad \text{or} \quad x + y + 1 = 0$$

(ii) METHOD 1: TEST POINT OF INTERSECTION

The curve intersects with l when:

$$x^4 - 2x^2 - x = -x - 1$$

$$x^4 - 2x^2 + 1 = 0$$

$$(x^2 - 1)^2 = 0$$

$$x = \pm 1 \quad \textcircled{1} \text{ locates points of intersection}$$

Since $x=1$ gives the original intersection point, the other must be $x=-1$.

$$\text{When } x=-1, \frac{dy}{dx} = -4 + 4 - 1 = -1 \quad \left. \vphantom{\frac{dy}{dx}} \right\} \textcircled{1} \text{ shows that gradients are equal}$$

$\therefore$ When $x=-1$, l intersects with the curve & their gradients are equal, so it is a tangent here as well.

$$\text{When } x=-1, y = 1 - 2 + 1$$

$$= 0 \quad \therefore \text{Coordinates are } (-1, 0).$$

$\textcircled{1}$ full coordinates

METHOD 2: EQUATE GRADIENTS

If l is tangent to the curve elsewhere, there must be another point with the same gradient.

$$\text{ie. } \frac{dy}{dx} = -1$$

$$4x^3 - 4x - 1 = -1$$

$$4x^3 - 4x = 0$$

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x = 0 \text{ or } \pm 1$$

$x=1$ gives the original intersection point.

When $x=0$, $y=0$.

$$\text{When } x=-1, y = 1 - 2 + 1 = 0$$

ie. The curve has gradient -1 at points $(0,0)$ and $(-1,0)$.

Test these coordinates with line l ($y = -x - 1$):

$$0 \neq -0 - 1 \rightarrow (0,0) \text{ is not on line } l$$

$$0 = -(-1) - 1 \rightarrow (-1,0) \text{ is on line } l$$

$\therefore$ Line l is also tangent to the curve at $(-1,0)$.

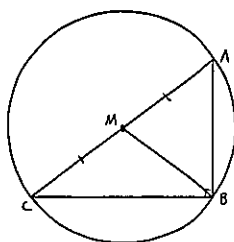
QUESTION 13

(a) METHOD 1: CIRCLE PROPERTIES

Since $\angle ABC$ is a right angle, it is the angle in a semicircle with diameter AC .

But M is the midpoint of AC , so it is the centre of the circle.

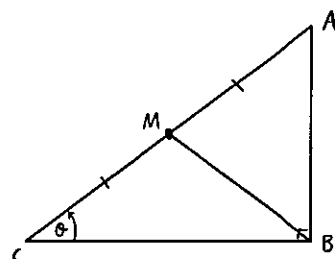
$\therefore BM = CM$ (radii)



METHOD 2: TRIGONOMETRY

Let $\angle ACB = \theta$.

$$\begin{aligned} \text{In } \triangle ABC, \quad \cos \theta &= \frac{BC}{AC} \\ &= \frac{BC}{2 \times CM} \quad (\text{since } M \text{ bisects } AC) \end{aligned}$$



$$\text{In } \triangle BCM, \quad \cos \theta = \frac{CM^2 + BC^2 - BM^2}{2 \times CM \times BC}$$

$$\frac{BC}{2 \times CM} = \frac{CM^2 + BC^2 - BM^2}{2 \times CM \times BC}$$

$$BC = \frac{CM^2 + BC^2 - BM^2}{BC}$$

$$BC^2 = CM^2 + BC^2 - BM^2$$

$$BM^2 = CM^2$$

$\therefore BM = CM$ ($BM > 0$ & $CM > 0$ as they are lengths)

METHOD 3: COORDINATE GEOMETRY

Since $\triangle ABC$ is right-angled at B , position B at the origin with A & C on the x & y axes respectively.

Let coordinates of A be $(x_1, 0)$

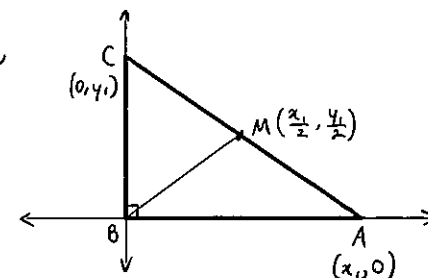
and C be $(0, y_1)$.

$\therefore M$ is at $(\frac{x_1}{2}, \frac{y_1}{2})$.

$$\begin{aligned} BM &= \sqrt{(\frac{x_1}{2} - 0)^2 + (\frac{y_1}{2} - 0)^2} \\ &= \sqrt{\frac{(x_1)^2}{4} + \frac{(y_1)^2}{4}} \end{aligned}$$

$$\begin{aligned} CM &= \sqrt{(\frac{x_1}{2} - 0)^2 + (\frac{y_1}{2} - y_1)^2} \\ &= \sqrt{\frac{(x_1)^2}{4} + \frac{(-y_1)^2}{4}} \\ &= \sqrt{\frac{(x_1)^2}{4} + \frac{(y_1)^2}{4}} \end{aligned}$$

$\therefore CM = BM$ as required.



METHOD 4 (triangle congruence)

Construct MN where N is the midpoint of BC .

$\therefore MN \parallel AB$ (interval joining midpoints of two sides in $\triangle ABC$ is parallel to the third side)

$\therefore \angle MNC = \angle ABC$ (corresponding angles on parallel lines are equal)
 $= 90^\circ$

$\angle MNC + \angle MNB = 180^\circ$ (angle sum of a straight line is 180°)

$$\angle MNB = 90^\circ$$

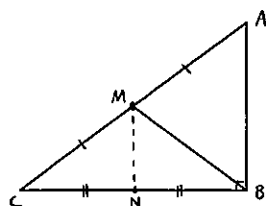
In $\triangle MNC$ & $\triangle MNB$: MN is common

$\angle MNC = \angle MNB$ (proven above)

$CN = BN$ (N bisects BC)

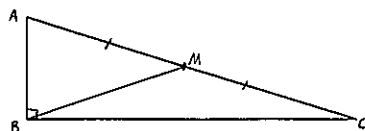
$\therefore \triangle MNC \equiv \triangle MNB$ (SAS)

$\therefore BM = CM$ (corresponding sides in congruent triangles are equal),
 as required.



Note that the information in the question does not guarantee that BM bisects $\angle ABC$, nor that BM is perpendicular to AC .

Consider the following possible $\triangle ABC$:

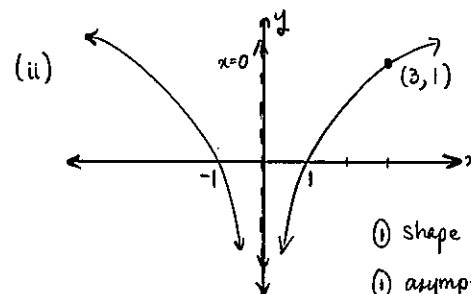


In this example, BM is clearly not perpendicular to AC , nor is $\angle ABM$ equal to $\angle CBM$ (though both of these were erroneously assumed by students).

$$(b) f(x) = \log_3(x^2)$$

Note that as $f(x) = f(-x)$, this is an even function.

$$(i) f(-1) = \log_3 1 \\ = 0$$



- ① shape & intercepts
- ① asymptote
- ① both branches

$$(c) \text{ RTP: } \frac{\tan^3 \theta + 1}{\tan \theta + 1} = \frac{1}{2} \sec^2 \theta (2 - \sin 2\theta)$$

METHOD 1: STANDARD IDENTITIES

$$\text{LHS} = \frac{\tan^3 \theta + 1}{\tan \theta + 1}$$

$$= \frac{(\tan \theta + 1)(\tan^2 \theta - \tan \theta + 1)}{\tan \theta + 1}$$

① sum of cubes factorisation

$$= \tan^2 \theta - \tan \theta + 1$$

$$= \sec^2 \theta - \tan \theta$$

① Pythagorean identity

$$= \sec^2 \theta - \frac{\sin \theta \cos \theta}{\cos^2 \theta}$$

$$= \sec^2 \theta - \frac{1}{2} \sin 2\theta \sec^2 \theta$$

① double angle identity

$$= \frac{1}{2} \sec^2 \theta (2 - \sin 2\theta) \text{ as required.}$$

METHOD 2: t-RESULTS

$$\text{Let } t = \tan \theta.$$

$$\therefore \sin 2\theta = \frac{2t}{1+t^2} \quad \text{and} \quad \sec^2 \theta = 1+t^2.$$

$$\text{LHS} = \frac{t^3+1}{t+1}$$

$$\text{RHS} = \frac{1}{2}(1+t^2)\left(2 - \frac{2t}{1+t^2}\right)$$

$$= (1+t^2) - t$$

$$= t^2 - t + 1$$

$$= \frac{(t+1)(t^2-t+1)}{t+1}$$

$$= \frac{t^3+1}{t+1} \quad \text{as required.}$$

$$(d) \quad x-2y + \sqrt{x+y} = \frac{1}{2}\sqrt{6}$$

$$x-2y + \sqrt{x+y} = \sqrt{\frac{6}{4}}$$

$$(x-2y) + \sqrt{x+y} = \sqrt{\frac{3}{2}}$$

By comparing rational & irrational components, } ① working

$$x-2y = 0 \quad \text{--- ①}$$

$$x+y = \frac{3}{2} \quad \text{--- ②}$$

$$\text{②} - \text{①} : y - (-2y) = \frac{3}{2} - 0$$

$$3y = \frac{3}{2}$$

$$y = \frac{1}{2}$$

$$\therefore x = 1$$

} ① final values

QUESTION 14

$$(a) \quad E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$(i) \quad \text{First 5 terms} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

$$(ii) \quad E'(x) = 0 + 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \dots$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$= E(x) \quad \text{as required.}$$

$$(b) \quad \cos 2x + \sqrt{3} \sin 2x = \sqrt{2} \quad \{0^\circ \leq x \leq 360^\circ\}$$

$$\text{Let } \cos 2x + \sqrt{3} \sin 2x = R \cos(2x - \alpha)$$

$$= R[\cos 2x \cos \alpha + \sin 2x \sin \alpha]$$

$$\cos 2x + \sqrt{3} \sin 2x = (R \cos \alpha) \cos 2x + (R \sin \alpha) \sin 2x$$

By comparison of coefficients,

$$R \cos \alpha = 1 \quad \text{--- ①}$$

$$R \sin \alpha = \sqrt{3} \quad \text{--- ②}$$

$$\text{①}^2 + \text{②}^2 : R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1 + 3$$

$$R^2(1) = 4$$

$$R = 2 \quad (R > 0)$$

① use of trigonometric expansion & formation of simultaneous equations

$\therefore \cos \alpha = \frac{1}{2}$ & $\sin \alpha = \frac{\sqrt{3}}{2}$ Note: do not accept responses that use
 $\rightarrow 0 < \alpha < 90^\circ$ $\frac{1}{2}$ and without accounting for domain of α

$\rightarrow \alpha = 60^\circ$ ① amplitude & phase shift (note that phase may
 $\therefore \cos 2\alpha + \sqrt{3} \sin 2\alpha = 2 \cos(2\alpha - 60^\circ)$ differ based on choice
of trigonometric expansion)

$$2 \cos(2\alpha - 60^\circ) = \sqrt{2}$$

$$\cos(2\alpha - 60^\circ) = \frac{1}{\sqrt{2}}$$

$$2\alpha - 60^\circ = -45^\circ, 45^\circ, 315^\circ, 405^\circ$$

$$2\alpha = 15^\circ, 105^\circ, 375^\circ, 465^\circ$$

$$\alpha = 7.5^\circ, 52.5^\circ, 187.5^\circ, 465^\circ$$
 ① all solutions

$$\begin{aligned} 0 < \alpha < 360 \\ 0 < 2\alpha < 720 \\ -60 < 2\alpha - 60 < 660 \end{aligned}$$

METHOD 2

$$\angle ACE + \angle CAE + \angle AEC = 180^\circ \text{ (angle sum of } \triangle ACE \text{ is } 180^\circ)$$

$$\angle ACE + \angle CAE + 90^\circ = 180^\circ$$

$$\angle ACE = 90^\circ - \angle CAE$$

But $\angle ABC = \angle ACE$ (angle in the alternate segment) ①

$$\angle ABC = 90^\circ - \angle CAE$$

$$\angle ACB = 90^\circ \text{ (angle in a semicircle is } 90^\circ) \text{ ①}$$

$$\angle BAC + \angle ACB + \angle ABC = 180^\circ \text{ (angle sum of } \triangle ABC \text{ is } 180^\circ) \text{ ①}$$

$$\angle BAC + 90^\circ + 90^\circ - \angle CAE = 180^\circ$$

$$\angle BAC = \angle CAE$$

$\therefore AC$ bisects $\angle EAB$, as required.

(c) METHOD 1

$$\angle AEC = 90^\circ \text{ (CE} \perp \text{AD)}$$

$$\angle ACB = 90^\circ \text{ (angle in a semicircle is } 90^\circ)$$

$$\therefore \angle AEC = \angle ACB$$

In $\triangle AEC$ & $\triangle ACB$:

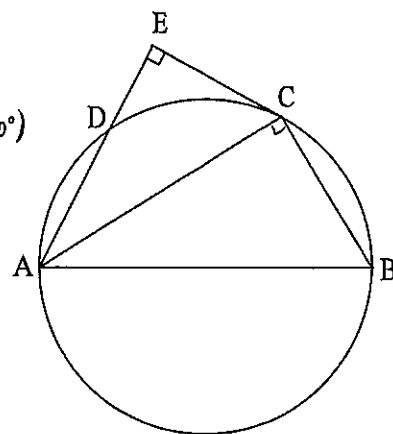
$$\angle AEC = \angle ACB \text{ (proven above)}$$

$$\angle ACE = \angle ABC \text{ (angle in the alternate segment)}$$

$$\therefore \triangle AEC \parallel \triangle ACB \text{ (equiangular)}$$

$$\therefore \angle CAE = \angle BAC \text{ (corresponding angles of similar triangles are equal)}$$

$\therefore AC$ bisects $\angle EAB$, as required.



(d)(i) $\angle PZS = 90^\circ$ ($AP \perp SZ$)

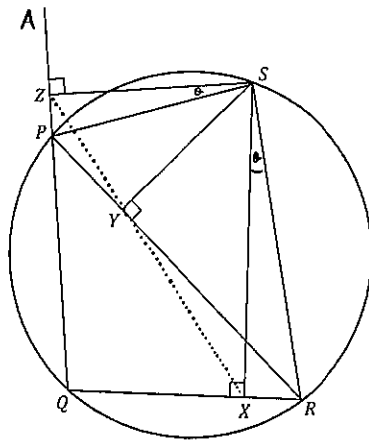
$$\angle SPQ = 90^\circ \text{ (PR} \perp \text{SY)} \quad \textcircled{1}$$

$$\therefore \angle SYP + \angle PZS = 180^\circ$$

$\therefore$ PQSR is a cyclic quadrilateral

(opposite angles are supplementary) ①

$\therefore P, S, Y$ & Z are concyclic, as required.



(ii) $\angle RXS = 90^\circ$ ($QR \perp SX$) ①

$$\therefore \angle RXS = \angle RYS$$

$\therefore \angle RXS$ & $\angle RYS$ are angles in the same segment

standing on the same arc RS of circle RXYZ

(note: not arc RS of circle PQRS)

$\therefore R, S, X$ & Y are concyclic, as required.